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REDUCED-RIPPLE CHARGE PUMP SYSTEMS AND METHODS

Abstract

An electrical device may include an electrical load that draws current, a charge pump that provides the current to the electrical load, and clock circuitry. The charge pump may include an amplifier, a first fly capacitor, a second fly capacitor, and multiple switches. A first set of the switches may couple the first fly capacitor between a supply and ground based on a first set of clock signals and between the amplifier and the electrical load based on a second set of clock signals. A second set of the switches may couple the second fly capacitor between the supply and ground based on the second set of clock signals and between the amplifier and the electrical load based on the first set of clock signals. The clock circuitry may alternate between generating the first set of clock signals and the second set of clock signals based on the current.

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Background/Summary

BACKGROUND

[0001] This disclosure relates to systems and methods of power management circuitry for semiconductor devices and, more particularly, to charge pumps for increasing voltage potential.

[0002] Integrated circuits are found in a vast array of electronics devices, including computers, handheld devices, wearable devices, vehicles, robotics, and more. An electronic device may be operated in various operational modes, such as a normal mode, a powered-off mode, and a reduced-power mode, among others. In some systems, circuit blocks designed to perform various functions may be designed to operate at different power supply levels (e.g., voltage levels). Power management circuitry may vary power supply levels delivered to the different circuit blocks.

[0003] Power management circuits sometimes include one or more power converter circuits that generate regulated voltage levels based on an input power supply signal. Such regulator circuits may employ different operations to regulate a respective voltage level. For example, a power management circuit may include a charge pump for providing an increased direct current (DC) voltage output. However, operation of these power circuits, such as a typical charge pump, could result in noise (e.g., switching harmonics, voltage deviations such as ripples) being introduced into the generated supply signals. Noise in the supply signals may be difficult to filter out and/or may undesirably reduce the accuracy of downstream system operations based on the signals generated.

SUMMARY

[0004] Integrated circuit systems may include multiple circuits to perform specific operations. The circuits may be fabricated on one or more substrates and may use different power (e.g., voltage) levels. Power Management Units (PMUs) may include multiple power converter circuits that generate regulated voltage levels for various power signals delivered to one or more loads. In some embodiments, such power converter circuits may be designed to generate voltage levels greater than (e.g., relative to a ground reference) that of an input reference voltage or less than a ground reference (e.g., negative voltage). For example, a charge pump may receive a supply voltage and provide an output voltage at a greater potential difference from the ground reference than the reference voltage.

[0005] However, typical charge pump circuits may generate ripples or other deviations in the output voltage and/or utilize large output capacitors to help mitigate the generated ripples. In overcoming the deficiencies of typical charge pump circuits, a reduced-ripple charge pump may provide an increased voltage level (e.g., relative to the input supply voltage) for a desired load without or with minimal ripple effects, with reduced output capacitor size, and/or reduced switching frequency. Thus, the reduced-ripple charge pump may provide for increased power efficiency, decreased footprint, and a cleaner (e.g., with less voltage variation) voltage output than a typical charge pump circuit.

[0006] In some embodiments, the reduced-ripple charge pump may include a current source (e.g., operational transconductance amplifier (OTA)) and two or more stages that alternate between phases according to a clock signal. Additionally, each stage may include at least one fly capacitor, switches (e.g., transistors) that operatively connect and disconnect the different leads of a fly capacitor to charge or discharge according to the phase.

[0007] As the clock signal switches the phases of the stages, the fly capacitor(s) of the first stage and the second stage alternate between being charged (e.g., between the ground voltage and the supply voltage) in the first phase and discharged (e.g., between the output node and the amplifier) in the second phase. During the second phase, the voltage differential (applied to the fly capacitor in the first phase) is added to the voltage output of an OTA to achieve the output voltage of the output node. However, the fly capacitor may not discharge immediately or entirely. Instead, current may be supplied from the OTA to match the current usage on the output node (e.g., by a load of the reduced-ripple charge pump), and as the fly capacitor is discharged (e.g., reducing its stored voltage differential), the OTA may increase its voltage level output to maintain the output voltage of the output node. When the amplifier reaches its maximum voltage output (e.g., for the given supply voltage) or within a buffer of the maximum voltage output, the stages may alternate such that the second stage feeds the output node while the first stage charges.

[0008] The rate of discharge of the fly capacitors may depend on the current draw of the load. As such, in some embodiments, the clock signal may have a preset frequency or be adjustable, such as via a clock controller, based on the current draw of the load. The clock controller may estimate, measure, or otherwise assume (e.g., based on an operating mode of the electronic device, the load) a current draw of the load and adjust the clock frequency based thereon. Additionally or alternatively, the clock controller may include a comparator to monitor the voltage differential of the fly capacitors. As such, switching between phases may be triggered based on how quickly the fly capacitor discharges. In other words, the reduced-ripple charge pump may self-regulate the switching frequency, which may increase efficiency. As such, an electronic device may include a reduced-ripple charge pump, such as in a PMU, for providing an increased (e.g., relative to the supply voltage) output voltage without or with reduced voltage deviations (e.g., ripples, ringing), while having a reduced footprint and/or increased power efficiency, as compared to a conventional charge pump circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings described below.

[0010] FIG. **1** is a schematic block diagram of an electronic device including a power management unit (PMU), in accordance with an embodiment;

[0011] FIG. **2** is a front view of a mobile phone representing an example of the electronic device of FIG. **1**, in accordance with an embodiment;

[0012] FIG. **3** is a front view of a tablet device representing an example of the electronic device of FIG. **1**, in accordance with an embodiment;

[0013] FIG. **4** is a front view of a notebook computer representing an example of the electronic device of FIG. **1**, in accordance with an embodiment;

[0014] FIG. **5** are front and side views of a watch representing an example of the electronic device of FIG. **1**, in accordance with an embodiment;

[0015] FIG. **6** is a schematic diagram of a reduced-ripple charge pump of the PMU of FIG. **1**, in accordance with an embodiment;

[0016] FIG. **7** is a timing diagram of power characteristics of the reduced-ripple charge pump of FIG. **6**, in accordance with an embodiment;

[0017] FIG. **8** is a schematic diagram of a reduced-ripple negative charge pump of the PMU of FIG. **1**, in accordance with an embodiment;

[0018] FIG. **9** is a timing diagram of power characteristics of the reduced-ripple negative charge pump of FIG. **8**, in accordance with an embodiment;

[0019] FIG. **10** is a schematic diagram of clock circuitry coupled to a portion of the reduced-ripple charge pump of FIG. **6** or the reduced-ripple negative charge pump of FIG. **8**, in accordance with an embodiment;

[0020] FIG. **11** is a timing diagram of power characteristics of the clock circuitry and portion of the reduced-ripple charge pump of FIG. **10**, in accordance with an embodiment; and

[0021] FIG. **12** is a flowchart of an example process for operating the reduced-ripple charge pump of FIG. **6** or the reduced-ripple negative charge pump of FIG. **8**, in accordance with an embodiment.

DETAILED DESCRIPTION

[0022] When introducing elements of various embodiments of the present disclosure, the articles "a," "an," and "the" are intended to mean that there are one or more of the elements. The terms "including" and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to "some embodiments," "embodiments," "one embodiment," or "an embodiment" of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, the phrase A "based on" B is intended to mean that A is at least partially based on B. Moreover, the term "or" is intended to be inclusive (e.g., logical OR) and not exclusive (e.g., logical XOR). In other words, the phrase A "or" B is intended to mean A, B, or both A and B.
[0023] Electronic devices such as computer systems may include multiple circuits to perform specific operations. The circuits may be fabricated on one or more substrates and may be designed for use at different voltage levels. In general, an electronic may receive power at one or more supply voltages, such as from a battery, power adapter connected to a wall outlet or car battery, etc. Power Management Units (PMUs) may include multiple power converter circuits that generate regulated voltage levels for various power supply signals delivered to one or more loads. In some scenarios, it may be desired that a power converter circuit generate a voltage level greater than (e.g., relative to a ground reference voltage) that of an input supply voltage or less than a ground voltage (e.g., negative voltage).
[0024] As such, in some embodiments, a charge pump may be utilized to receive a supply voltage and provide an output voltage greater than the supply voltage. However, typical charge pump circuits may generate ripples (e.g., ringing) in the output voltage due to the switch-pulsed discharge of fly capacitors. Such ripples or other signal distortions may be exacerbated at higher current draws of the load. For example, the maximum load of a typical charge pump may be proportional to the size of the ripple effect, the switching frequency of the charge pump, and the size of the output capacitor. However, increasing the switching frequency may increase power consumption and/or introduce additional avenues for error. Moreover, increasing the size of the output capacitor may take up valuable space within a PMU and/or the electronic device as well as increase the costs associated with manufacturing the charge pump. To help overcome such deficiencies in typical charge pump circuits, a reduced-ripple charge pump may provide an increased voltage level (e.g., relative to the input supply voltage) for a desired load without or with minimal increase in switching frequency and/or output capacitor size. Thus, the reduced-ripple charge pump may provide for increased power efficiency, decreased footprint, and a cleaner (e.g., with less voltage variation) voltage output than a typical charge pump circuit.
[0025] In some embodiments, the reduced-ripple charge pump may include a current source (e.g., amplifier) and two or more stages, each having at least one fly capacitor, that alternate between phases according to a clock signal. Additionally, each stage may include switches (e.g., transistors) that operatively connect and disconnect the leads of a fly capacitor according to the phase. For example, the first and second leads of the fly capacitor of a first stage may be electrically coupled to a ground voltage and a supply voltage, respectively, during a first phase. Moreover, the first and second leads of the fly capacitor of the first stage may be electrically coupled to an amplifier, such as an operational transconductance amplifier (OTA), and an output node of the reduced-ripple charge pump, respectively, during a second phase. As should be appreciated, while discussed herein as utilizing an OTA, the reduced-ripple charge pump may utilize any suitable amplifier or other current source for supplying current over a range of voltages.
[0026] As the clock switches between phases, the fly capacitor of the first and second stages alternate between being charged (e.g., between the ground voltage and the supply voltage) in the first phase and discharged (e.g., between the output node and the amplifier) in the second phase. During the second phase, the voltage differential applied to the fly capacitor in the first phase) is added to the voltage output of the amplifier to achieve the output voltage of the output node.

However, the fly capacitor may not discharge immediately or entirely. Instead, current may be supplied from the amplifier to match the current usage on the output node (e.g., by a load of the reduced-ripple charge pump). As the fly capacitor is discharged (e.g., reducing its stored voltage differential), the amplifier may increase its voltage level output to maintain the output voltage of the output node. When the amplifier reaches its maximum voltage output (e.g., for the given supply voltage) or within a buffer of the maximum voltage output, the stages may alternate such that the second stage feeds the output node while the first stage charges. As should be appreciated, the buffer may be based on the maximum voltage output (e.g., the supply voltage) of the amplifier, a timing of the clock signal, a current draw of the load, the voltage of the output node, or a combination thereof.

[0027] As the amplifier provides a matched current draw of the output node to the fly capacitor in the second phase, the discharge time of the fly capacitor may vary based on the size of the fly capacitor and the current draw of the load on the output node. In other words, at lower current draws, the fly capacitors may discharge slower in the second phase than at higher current draws. As such, in some embodiments, the frequency of the clock signal may be adjusted to maintain the output voltage based on the current draw of the load. For example, at a lower current draw, there may be additional time before the combined output of the decreasing voltage differential of the fly capacitor and the increasing (e.g., until the supply voltage or a buffer amount therefrom is reached) output voltage of the amplifier can no longer sustain the output voltage. As such, the length of time between phase switches may be decreased at lower current draws.

[0028] In some embodiments, the clock signal may have a preset frequency or be controllable, such as via a clock controller. The clock controller may estimate, measure, or otherwise assume (e.g., based on an operating mode of the electronic device, the load) a current draw of the load and adjust the clock frequency based thereon. Additionally or alternatively, the clock controller may include a comparator to monitor the voltage differential of the fly capacitors. As such, switching between phases may be triggered based on how quickly the fly capacitor discharges. Moreover, in some embodiments, the clock controller may be integral with the reduced-ripple charge pump such that the reduced-ripple charge pump is considered to self-clock based on the current draw of the load. As such, an electronic device may include a reduced-ripple charge pump, such as in a PMU, for providing an increased (e.g., relative to the supply voltage) output voltage without or with reduced voltage deviations (e.g., ripples, ringing), reduced footprint, and/or increased power efficiency, as compared to a conventional charge pump circuit.

[0029] With the foregoing in mind, FIG. **1** is a block diagram of an example electronic device **10** that may utilize the techniques of the present disclosure. In some embodiments, the electronic device **10** may include an electronic display **12**, one or more input devices **14**, one or more input/output (I/O) ports **16**, a processor core complex **18** having one or more processor(s) or processor cores, local memory **20**, a main memory storage device **22**, a network interface **24**, a power source **26** (e.g., power supply), and a power management unit (PMU) **28**. As should be appreciated, the electronic device **10** may be any suitable electronic device, such as a computer, a mobile phone, a portable media device, a tablet, a television, a virtual-reality headset, a wearable device such as a watch, a vehicle and/or vehicle dashboard, or the like. Indeed, FIG. **1** is intended to represent one example of a particular implementation and is intended to illustrate the types of components that may be present in the electronic device **10**. The various components described in FIG. **1** may include hardware elements (e.g., circuitry), software elements (e.g., a tangible, non-transitory computer-readable medium storing executable instructions), or a combination of both hardware and software elements. It should be noted that the various depicted components may be combined into fewer components or separated into additional components. For example, the local memory **20** and the main memory storage device **22** may be included in a single component.

[0030] The processor core complex **18** is operably coupled with local memory **20** and the main memory storage device **22**. Thus, the processor core complex **18** may execute instructions stored in local memory **20** or the main memory storage device **22** to perform operations, such as generating or transmitting image data to display on the electronic display **12**. As such, the processor core complex **18** may include one or more general purpose microprocessors, one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), or any combination thereof.

[0031] In addition to program instructions, the local memory **20** or the main memory storage device **22** may store data to be processed by the processor core complex **18**. Thus, the local memory **20** and/or the main memory storage device **22** may include one or more tangible, non-transitory, computer-readable media. For example, the local memory **20** may include random access memory (RAM) and the main memory storage device **22** may include read-only memory (ROM), rewritable non-volatile memory such as flash memory, hard drives, optical discs, or the like.

[0032] The network interface **24** may communicate data with another electronic device or a network. For example, the network interface **24** (e.g., a radio frequency system) may enable the electronic device **10** to communicatively couple to a personal area network (PAN), such as a Bluetooth network, a local area network (LAN), such as an 802.11x Wi-Fi network, or a wide area network (WAN), such as a 4G, Long-Term Evolution (LTE), or 5G cellular network. The power source **26** may provide electrical power to one or more components in the electronic device **10**, such as the processor core complex **18** or the electronic display **12**. Thus, the power source **26** may include any suitable source of energy, such as a rechargeable lithium polymer (Li-poly) battery or an alternating current (AC) power converter. One or more PMU(s) **28** may help distribute power to various circuitries of the electronic device **10**. As should be appreciated, features described herein as relating to a PMU **28** may apply to multiple PMUs **28**. Moreover, some descriptions included herein may apply to systems with one PMU **28** and/or to systems with multiple PMUs **28**. Furthermore, PMUs **28** may include different components relative to each other, for example some PMUs **28** may include regulators while other PMUs **28** may not include regulators, as is described further herein. Additionally, in some embodiments, a PMUs **28** may be integrated with and/or be coupled to one or more chip packages (e.g., as one or more loads).

[0033] The I/O ports **16** may enable the electronic device **10** to interface with other electronic devices. For example, when a portable storage device is connected, the I/O port **16** may enable the processor core complex **18** to communicate data with the portable storage device. The input devices **14** may enable user interaction with the electronic device **10**, for example, by receiving user inputs via a button, a keyboard, a mouse, a trackpad, a touch sensing, or the like. The input device **14** may include touch-sensing components (e.g., touch control circuitry, touch sensing circuitry) in the electronic display **12**. The touch-sensing components may receive user inputs by detecting occurrence or position of an object touching the surface of the electronic display **12**.

[0034] The electronic device **10** may take any suitable form. One example of the electronic device **10** in the form of a handheld device **10**A is shown in FIG. **2**. The handheld device **10**A may be a portable phone, a media player, a personal data organizer, a handheld game platform, or the like. For illustrative purposes, the handheld device **10**A may be a smartphone, such as any IPHONE® model available from Apple Inc.

[0035] The handheld device **10**A includes an enclosure **30** (e.g., housing). The enclosure **30** may protect interior components from physical damage or shield them from electromagnetic interference, such as by surrounding the electronic display **12**. The electronic display **12** may display a graphical user interface (GUI) **32** having an array of icons. When an icon **34** is selected either by an input device **14** or a touch-sensing component of the electronic display **12**, an application program may launch.

[0036] The input devices **14** may be accessed through openings in the enclosure **30**. The input devices **14** may enable a user to interact with the handheld device **10**A. For example, the input devices **14** may enable the user to activate or deactivate the handheld device **10**A, navigate a user interface to a home screen, navigate a user interface to a user-configurable application screen, activate a voice-recognition feature, provide volume control, or toggle between vibrate and ring modes.

[0037] Another example of a suitable electronic device **10**, specifically, a tablet device **10**B, is shown in FIG. **3**. The tablet device **10**B may be any IPAD® model available from Apple Inc. A further example of a suitable electronic device **10**, specifically a computer **10**C, is shown in FIG. **4**. For illustrative purposes, the computer **10**C may be any MACBOOK® or IMAC® model available from Apple Inc. Another example of a suitable electronic device **10**, specifically a watch **10**D, is shown in FIG. **5**. For illustrative purposes, the watch **10**D may be any APPLE WATCH® model available from Apple Inc. As depicted, the tablet device **10**B, the computer **10**C, and the watch **10**D each also includes an electronic display **12**, input devices **14**, I/O ports **16**, and an enclosure **30**. The electronic display **12** may display a GUI **32**. Here, the GUI **32** shows a visualization of a clock. When the visualization is selected either by the input device **14** or a touch-sensing component of the electronic display **12**, an application program may launch, such as to transition the GUI **32** to presenting the icons **34** discussed in FIGS. **2** and **3**. Additionally, the electronic device may include one or more cameras **36**, such as for capturing images.

[0038] Referring back to FIG. **1**, circuitry of the electronic device **10**, such as those illustrated in FIG. **1**, may be fabricated on one or more substrates and may employ different power source **26** voltage levels. The PMU **28** may control and/or monitor the delivery of electrical power from the power source **26**, which may be done via on one or more voltage regulator circuits that generate regulated voltage levels to be delivered to the various circuitries of the electronic device **10**. For example, the PMU **28** may adjust the electrical power delivered to one or more domains, such as an analog domain, a digital domain, or the like. The PMU **28** may adjust the voltage levels based on operational modes instructed by the processor core complex **18** and/or based on expected or desired energy consumption levels of one or more different components or systems of the electronic device **10**. Such voltage regulator circuits may employ passive circuit elements (e.g., inductors, capacitors) and/or active circuit elements (e.g., transistors, diodes). Different types of voltage regulator circuits may be employed based on power usage of load circuits, available circuit area, and the like.

[0039] With the foregoing in mind, in some embodiments, a voltage regulator circuit of a PMU **28** may be or include a reduced-ripple charge pump **40** (e.g., charge pump circuitry), such as in the example schematic diagram of FIG. **6**. In some embodiments, the reduced-ripple charge pump **40** may include a current source, such as an operational transconductance amplifier (OTA) **42** and two or more stages **44** (e.g., a first stage **44**A and a second stage **44**B, cumulatively **44**). Each stage **44** may include one or more fly capacitors **46** and a set of switches **48** (e.g., transistors) that operatively connect and disconnect the fly capacitors **46** to a load **50** and the OTA **42** or to a ground reference **52** and a supply voltage **54**. Furthermore, in some embodiments, the connections of the fly capacitors **46** may alternate according to a clock signal **56**. Additionally, in some embodiments, an output capacitor **58** may provide smoothing/stability to the output voltage **60** (VOUT). However, by reducing or eliminating ripple production through the techniques discussed herein the traditional reliance on an output capacitor **58** to reduce ripple is broken, such that the output capacitor **58** may be reduced in size depending on implementation and/or load **50**.

[0040] The stages **44** of the reduced-ripple charge pump **40** may alternate between a first phase **62** (e.g., charging phase) and a second phase **64** (e.g., discharging phase) to charge and discharge, respectively, the fly capacitors **46** of opposite stages **44**. For example, when a stage **44** (e.g., first stage **44**A) is in the first phase **62**, a first lead **66**A of the fly capacitor **46** may be electrically coupled to the supply voltage **54** and a second lead of the fly capacitor **46** may be coupled to the ground reference **52**. Moreover, when a stage **44** (e.g., second stage **44**B) is in the second phase **64**, the first lead **66**A may be electrically coupled to the load **50** (or otherwise an output node of the reduced-ripple charge pump **40**) and the second lead **66**B may be coupled to the OTA **42**. While the first stage **44**A is in the first phase **62**, the second stage **44**B may be in the second phase **64** and vice versa. As should be appreciated, while discussed herein as utilizing an OTA **42**, the reduced-ripple charge pump **40** may utilize any suitable amplifier or other current source for supplying current over a range of voltages. Furthermore, while the OTA is depicted as utilizing the supply voltage **54** and the output voltage **60** as inputs, as should be appreciated, the voltages provided to the OTA **42** may be different. For example, the reference voltage supplied to the OTA **42** may or may not be the same as the supply voltage **54** used for charging the fly capacitors **46**.

[0041] By charging a fly capacitor **46** and coupling the charged fly capacitor **46** between the load **50** and the OTA **42**, the voltage differential of the charged fly capacitor **46** may be added to the voltage output of the OTA **42** to provide the output voltage **60** with a boosted voltage above that of the supply voltage **54**. To help illustrate, FIG. **7** is a timing diagram **70** of the power characteristics of the reduced-ripple charge pump **40** over time **72**. For example, the clock signal **56** may have change points **74** to switch the stages **44** between the different phases (e.g., first phase **62** and second phase **64**), while the output voltage stays constant regardless of phase or the draw of the load current **76**. The voltage differentials **78** (e.g., first stage voltage differential **78**A and second stage voltage differential **78**B, cumulatively **78**) of the fly capacitors **46** alternate (e.g., cycle) between charging and discharging at the change points **74** (e.g., according to the clock signal **56**).

[0042] When a stage **44** is in the second phase **64**, the voltage differential **78** is added to the OTA voltage output **80** to achieve the output voltage **60**. Current may be supplied from the OTA **42** to the fly capacitor **46** to match the current draw of the load **50**. Depending on the load current **76**, the fly capacitor **46** may discharge slower or faster. For example, after a load increase **82**, the voltage differential **78** for the fly capacitor **46** in the second phase **64** may decrease more rapidly. To maintain the constant output voltage **60** as the fly capacitor **46** is discharged (e.g., reducing its stored voltage differential **78**), the OTA **42** may increase the OTA voltage output **80** such that the sum of the OTA voltage output **80** and the voltage differential **78** of the fly capacitor **46** in the second phase **64** is maintained at the desired output voltage **60**. By utilizing controlled releases of the charge stored on the fly capacitors **46** in the second phase **64**, ripples in the output voltage **60** may be avoided that would otherwise be present due, at least in part, to the immediate discharges in traditional charge pumps.

[0043] In some embodiments, the clock signal **56** may be controlled such that when the OTA **42** reaches its maximum voltage output (e.g., for the given supply voltage **54** or other reference voltage) or within a buffer of the maximum voltage output of the OTA **42**, a change point **74** occurs, alternating the phases of the stages **44**. As should be appreciated, the buffer may be based on the maximum voltage output (e.g., the supply voltage) of the OTA **42**, a timing of the clock signal **56**, the load current **76** of the load **50**, the output voltage **60**, or a combination thereof. Additionally, as the output voltage **60** is a sum of the OTA voltage output **80** and the voltage differential **78**, the increase in voltage, relative to the supply voltage **54**, provided by a reduced-ripple charge pump **40** with two stages **44** may be set between x1 and x2 the supply voltage **54**. However, as should be appreciated, additional stages **44** may be implemented (e.g., in series) to increase the voltage multiplier output of the reduced-ripple charge pump **40**.

[0044] As discussed above, the reduced-ripple charge pump **40** may be utilized to generate voltages above the supply voltage **54**. In some embodiments, it may be desired to generate negative voltages. As such, in a similar manner as the reduced-ripple charge pump **40**, a reduced-ripple negative charge pump **90** (e.g., charge pump circuitry) may be employed in two or more stages **44**, as in the example schematic diagram of FIG. **8**. However, in the example implementation of the reduced-ripple negative charge pump **90**, the electrical connections of the OTA **42** and the load **50** are swapped, relative to the stages **44**. For example, to generate negative output voltages **60**, the OTA **42** may “push” charge in the opposite direction (e.g., relative to the leads **66** of the fly capacitor **46**) of the voltage differential **78** of a charged fly capacitor **46** to force a negative voltage on the other side of the fly capacitor **46**. Furthermore, in some embodiments, the OTA **42** may be a push-pull OTA or other amplifier with current sinking capability.

[0045] As with the reduced-ripple charge pump **40**, during the first phase **62** the first lead **66**A may be coupled to a supply voltage **54** and the second lead **66**B may be coupled to a ground reference **52**. However, in the second phase **64** of the reduced-ripple negative charge pump **90**, the first lead **66**A may be coupled to the OTA **42** (instead of the load **50**) and the second lead **66**B may be coupled to the load **50** (instead of the OTA **42**). By increasing the positive charge on the first lead **66**A of the fly capacitor **46** with the OTA **42**, a corresponding negative charge develops on the second lead **66**B coupled to the load **50** providing a negative output voltage **60**.
[0046] To help illustrate, FIG. **9** is a timing diagram **96** of power characteristics of the reduced-ripple negative charge pump **90** over time **72**. As should be appreciated, the fly capacitors **46** of the first stage **44**A (e.g., voltage differential **78**A) and the second stage **44**B (e.g., voltage differential **78**B) are charged and discharged in the same manner as the reduced-ripple charge pump **40** providing positive voltages. However, the OTA voltage output **80** starts at a positive voltage at the beginning of the second phase **64** and decreases as the fly capacitor **46** discharges. Indeed, as the voltage differential **78** decreases, less positive charge on the first lead **66**A is needed to maintain the negative charge on the second lead **66**B of the fly capacitor **46** and, therefore, maintain the constant negative voltage of the output voltage **60**.
[0047] As the voltage differential **78** is effectively directed in the opposite direction of the OTA **42**, the output voltage **60** may be the sum of the negative of the voltage differential **78** and the OTA voltage output **80**. For example, if the supply voltage is 10 volts and the OTA **42** supplies 5 volts during the second phase **64**, the output voltage may be −5 volts. Furthermore, while the OTA **42** may provide a positive voltage that decreases with the voltage differential **78**, the OTA **42** may sink the current equivalent to the load current **76**. When the OTA **42** reaches the ground reference **52** or within a buffer thereof, the clock signal may alternate the phases of the stages **44** so that the output voltage **60** is maintained at the desired negative voltage.
[0048] As discussed above, the OTA **42** provides (or sinks) current to match the load current **76** to the fly capacitor **46** in the second phase **64** to “push” or “pull” current to/from the load **50** from the fly capacitor **46**. Furthermore, the OTA **42** either increases the OTA voltage output **80** (e.g., in the case of a positive output voltage **60**) or decreases the OTA voltage output **80** (e.g., in the case of a negative output voltage **60**) according to the discharge of the fly capacitor **46**. Moreover, the discharge time of the fly capacitor may vary based on the size of the fly capacitor and the load current **76**. In other words, at lower current draws, the fly capacitors **46** may discharge slower in the second phase **64** than at higher current draws. As such, in some embodiments, the frequency of the clock signal **56** may be adjusted to maintain the output voltage **60** based on the load current **76** of the load **50**. For example, at a lower current draw, there may be additional time before the combined output of the decreasing voltage differential **78** of the fly capacitor **46** and the increasing (e.g., or decreasing in the case of the reduced-ripple negative charge pump **90**) OTA voltage output **80** can no longer sustain the desired output voltage **60**. As such, the length of time between phase switches may be decreased at lower current draws.
[0049] In some embodiments, the clock signal **56** may have a preset frequency or be controllable, such as via clock circuitry **100** (e.g., clock controller), as in the schematic diagram of FIG. **10**. The clock circuitry **100** may include clock logic **102** to estimate, measure, or otherwise assume (e.g., based on an operating mode of the electronic device **10**, the load **50**) a current draw of the load **50** and adjust the clock frequency based thereon. Furthermore, in some embodiments, the clock signal **56** may be regulated (e.g., controlled, generated) by one or more processors (e.g., processing circuitry), such as the processor core complex **18** or other processor(s). For example, the clock circuitry **100** may include or be coupled to one or more processors (e.g., processing circuitry) to control the switches **48**.
[0050] Additionally or alternatively, the clock circuitry **100** may include one or more comparators **104** to monitor the voltage differentials **78** of the fly capacitors **46**. As such, switching between phases may be triggered based on how quickly the fly capacitors **46** discharge. As should be

appreciated, the comparator(s) **104** may utilize any suitable comparator threshold depending on implementation. For example, a comparator **104** may compare the voltage differential **78** of a fly capacitor **46** to the supply voltage **54** (or other reference voltage) displaced by a voltage offset **106**. As should be appreciated, the comparator threshold, which may be obtained by the comparator **104** by the voltage offset **106** and/or other reference voltage, may be set based on the supply voltage **54**, the output voltage **60** and/or characteristics of the OTA **42**. Moreover, in some embodiments, the clock circuitry **100** may be integral with the reduced-ripple charge pump **40** (or reduced-ripple negative charge pump **90**) such that the reduced-ripple charge pump **40** is considered to self-clock based on the current draw (e.g., load current **76**) of the load **50**. Furthermore, in some embodiments, the switches **48** may be considered part of the clock circuitry **100**, part of the stages **44**, or both.

[0051] To help illustrate, FIG. **11** is a timing diagram **110** of power characteristics of the clock circuitry **100** in conjunction with the voltage differentials of the flying capacitors **46**. As discussed above, the voltage differentials **78** alternate discharge periods according to the current phase. When utilizing a self-clocking mode, the clock circuitry **100** may compare the voltage differentials **78** of the fly capacitors **46** to a comparator threshold **112** and trigger a change point **74** in the clock signal **56** when the voltage differential **78** reaches the comparator threshold **112**. As discussed above, the rate of discharge of the fly capacitors **46** may increase in response to an increase in the load current **76**. When utilizing the self-clocking mode, the clock signal **56** may likewise change frequency in response to the load current **76**. In the illustrated example, in response to the load increase **82** the clock signal frequency increases.

[0052] FIG. **12** is a flowchart of an example process for operating the reduced-ripple charge pump **40** or the reduced-ripple negative charge pump **90**. In general, a first fly capacitor **46** may be coupled between a supply voltage **54** and a ground reference **52**, charging the first fly capacitor **46** with a voltage differential (process block **122**). Additionally, the first fly capacitor **46** may be disconnected from the supply voltage **54** and the ground reference **52** and electrically coupled between an amplifier (e.g., OTA **42**) and a load **50**, beginning a discharge period of the first fly capacitor **46** (process block **124**). While the first fly capacitor **46** is coupled between the amplifier and the load **50**, a second fly capacitor **46** may be coupled between the supply voltage **54** and the ground reference **52**, charging the second fly capacitor (process block **126**). Furthermore, while the first fly capacitor **46** is coupled between the amplifier and the load **50**, a voltage may be provided to the first fly capacitor **46** via the amplifier such that a sum of the voltage differential **78** of the first fly capacitor and the output of the amplifier (e.g., OTA voltage output **80**) is equal to a desired voltage (e.g., output voltage **60**) at the load **50** throughout the discharge period (process block **128**). The charging and discharging periods of the first and second fly capacitors **46** may be alternated (process block **130**).

[0053] As such, an electronic device **10** may include a reduced-ripple charge pump **40** (or reduced-ripple negative charge pump **90**), such as in a PMU **28**, for providing an increased (e.g., relative to the supply voltage) output voltage (or negative voltage) without or with reduced voltage deviations (e.g., ripples, ringing), while having a reduced footprint (e.g., smaller or eliminated output capacitor **58**), and/or increased power efficiency, as compared to a conventional charge pump circuit.

[0054] The specific embodiments described above have been shown by way of example, and it should be understood that these embodiments may be susceptible to various modifications and alternative forms. It should be further understood that the claims are not intended to be limited to the particular forms disclosed, but rather to cover all modifications, equivalents, and alternatives falling within the spirit and scope of this disclosure.

[0055] Moreover, techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended

to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).
[0056] It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

Claims

1. An electrical device comprising: an electrical load configured to draw a current; a charge pump configured to provide the current to the electrical load at an output voltage, the charge pump comprising: an amplifier; a first fly capacitor; a second fly capacitor; a first set of switches configured to electrically couple the first fly capacitor between a supply voltage and a ground reference based on a first set of clock signals and to electrically couple the first fly capacitor between the amplifier and the electrical load based on a second set of clock signals; and a second set of switches configured to electrically couple the second fly capacitor between the supply voltage and the ground reference based on the second set of clock signals and to electrically couple the second fly capacitor between the amplifier and the electrical load based on the first set of clock signals; and clock circuitry configured to alternate between generating the first set of clock signals and the second set of clock signals based on the current.
2. The electrical device of claim 1, wherein the amplifier comprises an operational transconductance amplifier (OTA).
3. The electrical device of claim 1, wherein the output voltage is greater than the supply voltage.
4. The electrical device of claim 1, wherein the first fly capacitor is configured to be charged with a voltage differential when electrically coupled between the supply voltage and the ground reference.
5. The electrical device of claim 4, wherein the amplifier is configured to supply a voltage to the first fly capacitor while electrically coupled to the first fly capacitor such that output voltage is a sum of the voltage and the voltage differential.
6. The electrical device of claim 5, wherein the voltage differential is applied in a negative direction relative to the electrical load, the voltage is applied in a positive direction relative to the electrical load, and the output voltage is a negative voltage relative to the ground reference.
7. The electrical device of claim 4, wherein the clock circuitry is configured to alternate between generating the first set of clock signals and the second set of clock signals based a comparison between the voltage differential of the first fly capacitor and a threshold voltage, wherein a rate of change of the voltage differential is based on the current.
8. The electrical device of claim 1, wherein the clock circuitry is configured to increase a frequency of alternating between generating the first set of clock signals and the second set of clock signals based on an increase in the current.
9. The electrical device of claim 1, wherein the output voltage is a negative voltage relative to the ground reference.
10. Charge pump circuitry comprising: an amplifier; an electrical output configured to output an output voltage, wherein the output voltage is a negative voltage relative to a ground reference; a first fly capacitor configured to cycle between a charging phase and a discharging phase, wherein the first fly capacitor is configured to be electrically coupled between a supply voltage and the ground reference during the charging phase and electrically coupled between the amplifier and the

electrical output during the discharging phase; and a second fly capacitor configured to cycle between the charging phase and the discharging phase, wherein the second fly capacitor is configured to be electrically coupled between the supply voltage and the ground reference during the charging phase and electrically coupled between the amplifier and the electrical output during the discharging phase, wherein the first fly capacitor is configured to be operated in the charging phase while the second fly capacitor is operated in the discharge phase, and wherein the second fly capacitor is configured to be operated in the charging phase while the first fly capacitor is operated in the discharge phase.

11. The charge pump circuitry of claim 10, wherein the amplifier is configured to, while the first fly capacitor is in the discharge phase, provide a changing voltage to the first fly capacitor based on a voltage differential of the first fly capacitor.

12. The charge pump circuitry of claim 11, wherein the output voltage comprises a sum of the changing voltage of the amplifier and the voltage differential of the first fly capacitor in the discharge phase.

13. The charge pump circuitry of claim 10, wherein the first fly capacitor comprises a first lead and a second lead, wherein the first lead is configured to electrically couple to the supply voltage during the charging phase and the second lead is configured to electrically couple to the ground reference during the charging phase.

14. The charge pump circuitry of claim 13, wherein the first lead is configured to electrically couple to the amplifier during the discharging phase and the second lead is configured to electrically couple to the electrical output during the discharging phase.

15. The charge pump circuitry of claim 10, comprising a plurality of switches configured to: based on a first set of clock signals, electrically couple the first fly capacitor between the supply voltage and the ground reference and electrically couple the second fly capacitor between the amplifier and the electrical output; and based on a second set of clock signals, electrically couple the second fly capacitor between the supply voltage and the ground reference and electrically couple the first fly capacitor between the amplifier and the electrical output.

16. The charge pump circuitry of claim 15, comprising clock circuitry configured to generate the first set of clock signals and the second set of clock signals based on a current draw on the electrical output.

17. The charge pump circuitry of claim 15, wherein the plurality of switches comprises a plurality of transistors, and wherein the amplifier comprises an operational transconductance amplifier (OTA).

18. A non-transitory machine-readable medium comprising instructions, wherein, when executed by one or more processors, the instructions cause the one or more processors to perform operations or to control clock circuitry to perform the operations, wherein the operations comprise: providing a first set of clock signals to a plurality of switches of charge pump circuitry, wherein the first set of clock signals are configured to actuate the plurality of switches to electrically couple a first fly capacitor of the charge pump circuitry between a supply voltage and a ground reference and electrically couple a second fly capacitor of the charge pump circuitry between an amplifier of the charge pump circuitry and an electrical output of the charge pump circuitry; providing a second set of clock signals to the plurality of switches, wherein the second set of clock signals are configured to actuate the plurality of switches to electrically couple the second fly capacitor between the supply voltage and the ground reference and electrically couple the first fly capacitor between the amplifier and the electrical output; and alternating, at a frequency, between providing the first set of clock signals to the plurality of switches and providing the second set of clock signals to the plurality of switches, wherein the frequency is based on a load of the electrical output.

19. The non-transitory machine-readable medium of claim 18, wherein the operations comprise receiving an indication of the load, wherein the indication comprises a comparison between a reference voltage and a voltage differential of the first fly capacitor or the second fly capacitor.

20. The non-transitory machine-readable medium of claim 18, wherein an output voltage of the electrical output comprises a negative voltage relative to the ground reference.